# Mathematical Analysis 2

Multivariable derivatives 2: solutions

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1. Verify directly from the definition that the linear function $L(h_1, h_2) = 2(h_1 + h_2)$ is the total differential of $f(x, y) = x^2 + y^2$ at $(1, 1)$.

   **Solution.** The map $L$ is linear. Expanding the squares gives

   $$f(1 + h_1, 1 + h_2) - f(1, 1) - L(h) = h_1^2 + h_2^2 = \|h\|^2 .$$

   Hence $|f((1, 1) + h) - f(1, 1) - L(h)| / \|h\| = \|h\| \to 0$, which proves that $Df(1, 1) = L$.

2. Compute the directional derivatives of the following functions in the given directions at the specified points.

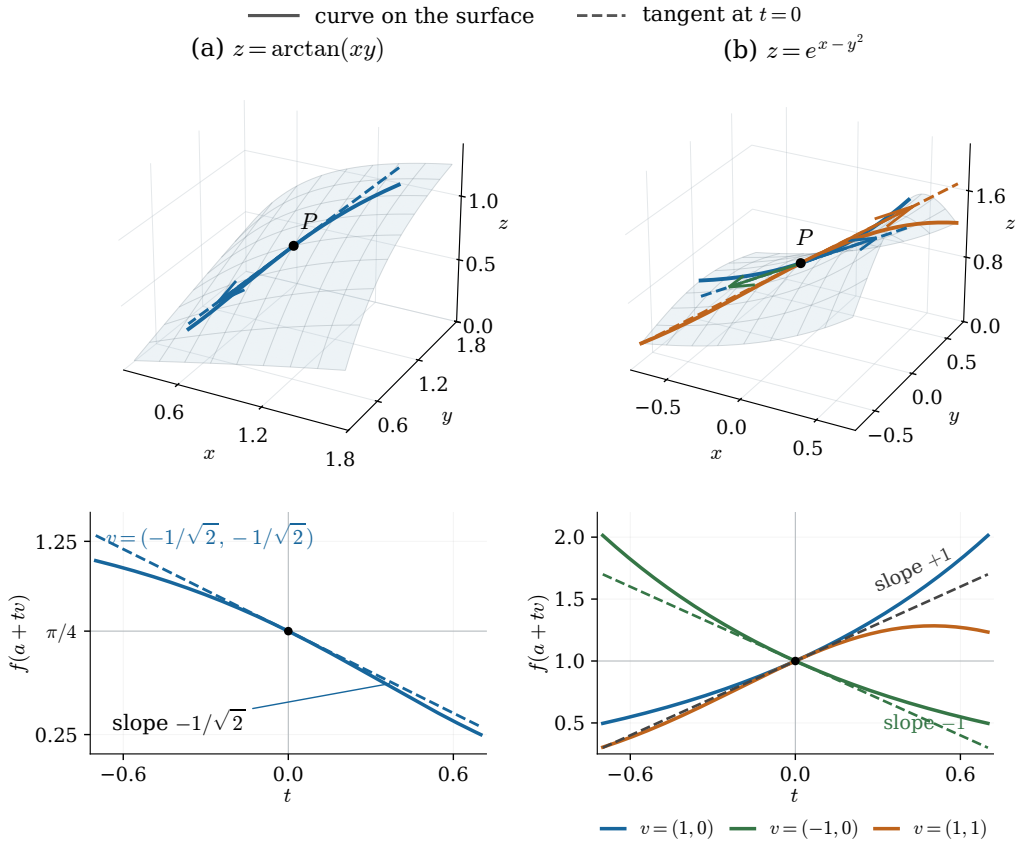
   (a) $f(x, y) = \arctan(xy)$ in the direction $v = (-1/\sqrt{2}, -1/\sqrt{2})$ at $a = (1, 1)$.

   (b) $f(x, y) = e^{x-y^2}$ in the directions $(1, 0)$, $(-1, 0)$, and $(1, 1)$ at $(0, 0)$.

   **Solution.** Both functions are smooth, so $D_v f(a) = \nabla f(a) \cdot v$. **(a)** $\nabla f(x, y) = (y, x)/(1 + x^2y^2)$, and therefore

   $$D_v f(1, 1) = (1/2, 1/2) \cdot (-1/\sqrt{2}, -1/\sqrt{2}) = -1/\sqrt{2}.$$

   **(b)** $\nabla f(x, y) = e^{x-y^2}(1, -2y)$, so $\nabla f(0, 0) = (1, 0)$. Taking its scalar product with each given direction gives 1, $-1$, and 1, respectively. The given directions are not normalized.




The upper plots show the surface paths $(a + tv, f(a + tv))$ through $P = (a, f(a))$; arrows indicate increasing $t$. The lower plots show the function value $f(a + tv)$ against $t$. The slope of the dashed tangent at $t = 0$ is $D_v f(a)$. In (b), the directions $(1, 0)$ and $(-1, 0)$ trace the same surface path in opposite directions. The blue and orange curves in the lower-right plot have the same tangent at $t = 0$, so both derivatives are 1. Slopes are measured per unit $t$, with each direction used as given.

3. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x,y) = \begin{cases} 1, & x = 0 \text{ or } y = 0, \\ 0, & x \neq 0 \text{ and } y \neq 0. \end{cases}$$

(a) Determine where $f$ is continuous.

(b) Compute the partial derivatives at points off the coordinate axes, at points on the axes other than $(0,0)$, and at $(0,0)$.

(c) Does $f$ have a total differential at $(0,0)$?

(d) Compute the directional derivative in the direction $(1,1)$ at points off the coordinate axes and at $(0,0)$.

**Solution. (a)** At $a = (a_1, a_2)$ with $a_1 a_2 \neq 0$, a ball of radius $\min\{|a_1|, |a_2|\}/2$ avoids both axes. Thus $f = 0$ on that ball and is continuous at $a$. Every point on an axis has value 1 but is approached by points off both axes with value 0; hence $f$ is discontinuous at every such point. **(b)** The partial derivatives are

| point | $\partial_x f$ | $\partial_y f$ |
|---|---|---|
| $xy \neq 0$ | 0 | 0 |
| $x = 0,\ y \neq 0$ | does not exist | 0 |
| $x \neq 0,\ y = 0$ | 0 | does not exist |
| $(0,0)$ | 0 | 0 |

The zero entries follow from local constancy off the axes and from $f = 1$ along each axis. At a nonzero point on an axis, moving in the perpendicular coordinate direction gives $(0-1)/t = -1/t$, which has no finite limit. **(c)** There is no total differential at $(0,0)$, because differentiability implies continuity. **(d)** Off the axes, local constancy gives $D_{(1,1)} f = 0$. At the origin, $[f(t,t) - f(0,0)]/t = -1/t$ for $t \neq 0$, so this directional derivative does not exist.

This is a counterexample to the claim that existence of all partial derivatives at a point implies total differentiability there. It also shows that partial derivatives alone guarantee neither continuity nor existence of all directional derivatives.

4. Consider the function

$$f(x,y) = \log(\sqrt{y+1} - x).$$

(a) Find and sketch its domain $D$. Is $f$ continuous?

(b) Compute $\nabla f(x,y)$ wherever it exists, and determine $\nabla f(0,0)$.

(c) Does $f$ have a total differential at $(0,0)$? If so, find it.

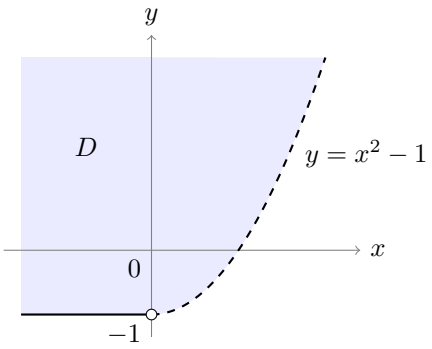
(d) Find the equation of the tangent plane at $(0,0)$.

(e) Use linear approximation to estimate $f(-0.04, 0.02)$.

**Solution. (a)** The square root requires $y \geq -1$ and the logarithm requires $\sqrt{y+1} - x > 0$. Thus the domain is

$$D = \{(x,y) : y \geq -1,\ x < \sqrt{y+1}\} = \{x < 0,\ y \geq -1\} \cup \{x \geq 0,\ y > x^2 - 1\}.$$

It lies above the solid half-line $y = -1$, $x < 0$ (included), and the dashed half-parabola $y = x^2 - 1$, $x \geq 0$ (excluded). The square root and logarithm are continuous wherever they are defined, so their composition is continuous on all of $D$, including its boundary points.




**(b)** At interior points of $D$, where $y > -1$ and $x < \sqrt{y+1}$, direct differentiation gives

$$\nabla f(x,y) = \left( -\frac{1}{\sqrt{y+1}-x}, \frac{1}{2\sqrt{y+1}\,(\sqrt{y+1}-x)} \right).$$

In particular, $\nabla f(0,0) = (-1, 1/2)$. At a boundary point $(x,-1)$, fix $x < 0$ and take a vertical step $t > 0$. Then $y = -1 + t$, so $t \to 0^+$ means $y \to -1^+$. Since $f(x,-1) = \log(-x)$, the one-sided difference quotient gives

$$\begin{aligned} \lim_{t\to 0^+} \frac{f(x,-1+t) - f(x,-1)}{t} &= \lim_{t\to 0^+} \frac{\log(\sqrt{t}-x) - \log(-x)}{t} \\ &= \lim_{t\to 0^+} \frac{\log(1+\sqrt{t}/(-x))}{t} = +\infty. \end{aligned}$$

To evaluate the last limit, factor the quotient:

$$\frac{\log(1+\sqrt{t}/(-x))}{t} = \underbrace{\frac{\log(1+\sqrt{t}/(-x))}{\sqrt{t}/(-x)}}_{\longrightarrow 1} \cdot \underbrace{\frac{1}{(-x)\sqrt{t}}}_{\longrightarrow +\infty}.$$

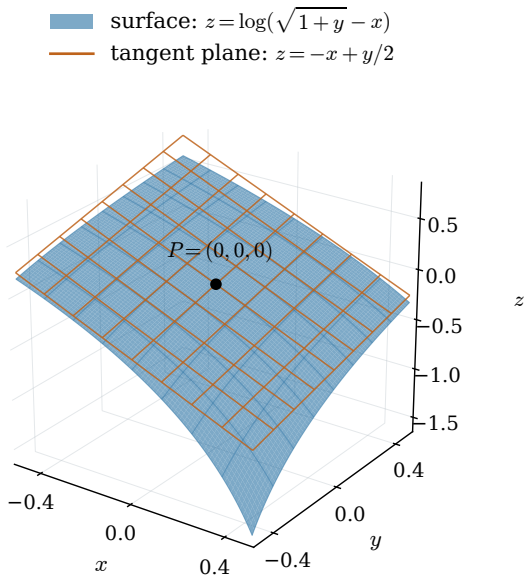
For the first factor, use the general limit below. Its form is $0/0$, so L'Hôpital's rule gives

$$\lim_{u\to 0} \frac{\log(1+u)}{u} = \lim_{u\to 0} \frac{1/(1+u)}{1} = 1.$$

Applying this with $u = \sqrt{t}/(-x) \to 0^+$ shows that the first factor tends to 1. The second is $+\infty$ because $-x > 0$ is fixed and $\sqrt{t} \to 0^+$. More explicitly, the first factor is at least $1/2$ for all sufficiently small $t > 0$, so the quotient is at least $1/(2(-x)\sqrt{t})$, which tends to $+\infty$. Thus $\partial_y f$ does not exist as a finite number on the boundary, so neither does the gradient. **(c)** The partial derivatives are continuous in a neighbourhood of $(0,0)$, so $f$ is differentiable there and $Df(0,0)h = \nabla f(0,0) \cdot h = -h_1 + h_2/2$. **(d)** Since $f(0,0) = 0$, the tangent plane $z = f(0,0) + Df(0,0)(x,y)$ is $z = -x + y/2$. **(e)** Substituting $h = (-0.04, 0.02)$ in the linear approximation gives $f(-0.04, 0.02) \approx 0 + 0.04 + 0.02/2 = 0.05$. For comparison, direct evaluation gives

$$f(-0.04, 0.02) = \log(\sqrt{1.02} + 0.04) \approx 0.0487430143.$$

Thus the linear approximation overestimates the actual value by approximately 0.001257, a relative error of about 2.58%.




The blue surface and the orange tangent plane meet at $P = (0, 0, 0)$ with the same slope in every direction.

5. Determine whether

$$f(x, y) = \frac{x + y}{x^2 + y^2} \ln(1 + xy)$$

can be extended to a neighbourhood of the origin so that it has a total differential at the origin.

**Solution.** No. If $r = \sqrt{x^2 + y^2} < 1$, then $|xy| \le r^2/2 < 1/2$, so $1 + xy > 0$ and the formula is defined away from the origin. The mean value theorem gives $|\ln(1 + u)| \le 2|u|$ for $|u| < 1/2$. Consequently,

$$|f(x, y)| \le \frac{2|x + y|\,|xy|}{x^2 + y^2} \le |x + y| \longrightarrow 0.$$

Thus any differentiable extension must have $f(0, 0) = 0$. On both coordinate axes the extended function is zero, so its two partial derivatives at the origin are zero. Any total differential would therefore be $L = 0$. However, along $(x, y) = (t, t)$ with $t > 0$,

$$\frac{f(t, t)}{\|(t, t)\|} = \frac{\ln(1 + t^2)}{\sqrt{2}\, t^2} \longrightarrow \frac{1}{\sqrt{2}} \neq 0.$$

This contradicts the defining limit for differentiability.

This is a counterexample to the claim that continuity together with existence of all partial derivatives at a point implies total differentiability there.

6. Find the directional derivatives of

$$f(x, y, z) = \sqrt[3]{x^3 + y^3 + z^3}$$

at $(0, 0, 0)$ in a direction $h = (h_1, h_2, h_3)$. Decide whether $f$ has a total differential at this point.

**Solution.** Since the real cube root satisfies $f(th) = tf(h)$ for all real $t$,

$$D_h f(0, 0, 0) = \lim_{t \to 0} \frac{f(th) - f(0)}{t} = \sqrt[3]{h_1^3 + h_2^3 + h_3^3}.$$

All partial derivatives are 1, so a total differential would have to be $L(h) = h_1 + h_2 + h_3$. But $D_{(1,1,0)} f(0) = \sqrt[3]{2} \neq 2$. Therefore no total differential exists.

This continuous function is a counterexample to the claim that existence of all directional derivatives at a point implies total differentiability there.

7. Compute the total differentials of the following functions. How can they be used to approximate the functions near a point $a \in \mathbb{R}^2$ or $a \in \mathbb{R}^3$, respectively?

   (a) $f(x, y) = e^{xy}$.

   (b) $f(x, y, z) = xy + yz + xz$.

   **Solution.** The gradients give the following candidates for the total differentials. We verify them directly by estimating the exact remainders. **(a)** $\nabla f(x, y) = e^{xy}(y, x)$. At $a = (a_1, a_2)$,

$$\begin{aligned} Df(a)h &= e^{a_1 a_2}(a_2 h_1 + a_1 h_2), \\ f(a+h) &\approx e^{a_1 a_2}\big(1 + a_2 h_1 + a_1 h_2\big), \\ R_a(h) &= e^{(a_1+h_1)(a_2+h_2)} - e^{a_1 a_2}\big(1 + a_2 h_1 + a_1 h_2\big). \end{aligned}$$

   Put $u = a_2 h_1 + a_1 h_2 + h_1 h_2$. Then

$$R_a(h) = e^{a_1 a_2}\big[(e^u - 1 - u) + h_1 h_2\big].$$

   The elementary exponential limit gives

$$\lim_{u \to 0} \frac{e^u - 1 - u}{u} = \lim_{u \to 0} \left( \frac{e^u - 1}{u} - 1 \right) = 0.$$

   By Cauchy–Schwarz and $2|h_1 h_2| \le h_1^2 + h_2^2$,

$$|u| \le \|a\| \, \|h\| + \frac{\|h\|^2}{2}, \qquad |h_1 h_2| \le \frac{\|h\|^2}{2}.$$

   Hence $u \to 0$ as $h \to 0$, and for $u \neq 0$,

$$\frac{|R_a(h)|}{\|h\|} \le e^{a_1 a_2} \left[ \left| \frac{e^u - 1 - u}{u} \right| \left( \|a\| + \frac{\|h\|}{2} \right) + \frac{\|h\|}{2} \right] \longrightarrow 0.$$

   If $u = 0$, the exponential remainder vanishes and $|R_a(h)| / \|h\| \le e^{a_1 a_2} \|h\| / 2 \to 0$ as well.

   **(b)** $\nabla f(x, y, z) = (y + z, x + z, x + y)$. At $a = (a_1, a_2, a_3)$,

$$\begin{aligned} Df(a)h &= (a_2 + a_3)h_1 + (a_1 + a_3)h_2 + (a_1 + a_2)h_3, \\ f(a+h) &\approx f(a) + (a_2 + a_3)h_1 + (a_1 + a_3)h_2 + (a_1 + a_2)h_3, \\ R_a(h) &= h_1 h_2 + h_2 h_3 + h_1 h_3. \end{aligned}$$

   Applying $2|h_i h_j| \le h_i^2 + h_j^2$ to each product gives

$$\frac{|R_a(h)|}{\|h\|} \le \frac{|h_1 h_2| + |h_2 h_3| + |h_1 h_3|}{\|h\|} \le \frac{h_1^2 + h_2^2 + h_3^2}{\|h\|} = \|h\| \longrightarrow 0.$$

   In both parts, $R_a(h)$ is the exact error:

$$f(a + h) = f(a) + Df(a)h + R_a(h).$$

   The bounds above prove $|R_a(h)| / \|h\| \to 0$ in each case. Thus the proposed linear maps are indeed the total differentials, and omitting $R_a(h)$ gives the stated linear approximations.

8. Show that, for small $x$ and $y$,

   (a) $(1+x)^m(1+y)^n \approx 1 + mx + ny$;

   (b) $\ln(1+x)\ln(1+y) \approx xy$.

   **Solution. (a)** Let $F(x,y) = (1+x)^m(1+y)^n$, with $m, n$ fixed. Near the origin both bases are positive and the partial derivatives are continuous. Moreover, $F(0,0) = 1$ and $\nabla F(0,0) = (m,n)$. Hence $F$ is differentiable at the origin, with $DF(0,0)(x,y) = mx + ny$. By definition,

$$\lim_{(x,y)\to(0,0)} \frac{F(x,y) - 1 - mx - ny}{\sqrt{x^2+y^2}} = 0,$$

   which justifies $F(x,y) \approx 1 + mx + ny$. **(b)** Since the derivative of ln at 1 is 1,

$$\lim_{t\to 0} \frac{\ln(1+t)}{t} = 1.$$

   Therefore, when $xy \neq 0$,

$$\frac{\ln(1+x)\ln(1+y)}{xy} = \frac{\ln(1+x)}{x}\,\frac{\ln(1+y)}{y} \longrightarrow 1 \qquad \text{as } (x,y)\to(0,0).$$

   This proves the approximation; when $xy = 0$, both expressions are zero.

9. Find the equation of the tangent plane to the graph of $f(x,y) = \arctan(x/y)$ at $(1,1,f(1,1))$.

   **Solution.** The function is smooth near $(1,1)$, $f(1,1) = \pi/4$, and $\nabla f(x,y) = (y,-x)/(x^2+y^2)$ for $y \neq 0$. Hence $\nabla f(1,1) = (1/2, -1/2)$, and the tangent plane is

$$z = \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{2}(y-1) = \frac{\pi}{4} + \frac{x-y}{2}.$$

10. The following function is known to be continuous:

$$f(x,y) = \begin{cases} (x^2+y^2)\sin\left(\dfrac{1}{x^2+y^2}\right), & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

    Compute its partial derivatives wherever they exist. Are they continuous at $(0,0)$? Find a candidate for the total differential at this point and check whether it is indeed the total differential.

    **Solution.** Write $s = x^2 + y^2$. For $s > 0$,

$$\partial_x f = 2x\sin(1/s) - \frac{2x}{s}\cos(1/s), \qquad \partial_y f = 2y\sin(1/s) - \frac{2y}{s}\cos(1/s).$$

    At the origin both partial derivatives are zero, since $f(t,0)/t = f(0,t)/t = t\sin(1/t^2)$ has absolute value at most $|t| \to 0$. Thus both partial derivatives exist everywhere. Neither partial derivative is continuous there: for $t_n = (2\pi n)^{-1/2}$, $\partial_x f(t_n, 0) = \partial_y f(0, t_n) = -2/t_n \to -\infty$. Since the partial derivatives at the origin are zero, the only possible total differential is $L = 0$. It is indeed the differential, because

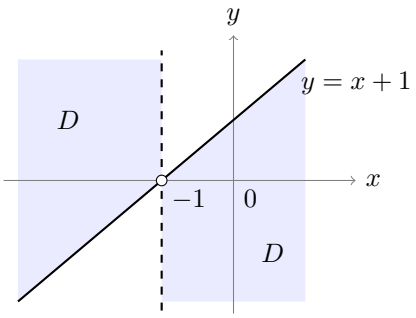
$$\frac{|f(h) - f(0) - L(h)|}{\|h\|} \leq \|h\| \longrightarrow 0.$$

    This is a counterexample to the claim that total differentiability at a point implies continuity of the partial derivatives there.

## Additional practice

11. Consider the function
$$f(x,y) = 4\sqrt{1 - \frac{y}{x+1}}.$$
(a) Find and sketch the domain of $f$.
(b) Compute the partial derivatives wherever they exist.
(c) Show that $f$ is differentiable at $(0,-3)$ and find its total differential there.
(d) Find the equation of the tangent plane to the graph of $f$ at $(0,-3,8)$.
(e) Estimate $f(-0.03,-3.01)$.

**Solution. (a)** The requirements $x \neq -1$ and $1 - y/(x+1) \geq 0$ give
$$D = \{x > -1,\ y \leq x+1\} \cup \{x < -1,\ y \geq x+1\}.$$
These are the two shaded regions below. The diagonal boundary $y = x+1$ is included except at $(-1,0)$; the vertical line $x = -1$ is excluded.




**(b)** Put $q = 1 - y/(x+1)$. At interior points of $D$, the chain rule gives
$$\partial_x f = \frac{2y}{(x+1)^2\sqrt{q}}, \qquad \partial_y f = -\frac{2}{(x+1)\sqrt{q}}.$$
At a boundary point $(x,y)$ with $y = x+1 = c \neq 0$, we have $f(x,y) = 0$ and
$$q(x+t,y) = \frac{t}{c+t}, \qquad q(x,y+t) = -\frac{t}{c}.$$
Along the respective allowed sides, both difference quotients have magnitude asymptotic to $4/(\sqrt{|c|}\sqrt{|t|}) \to \infty$. Thus neither partial derivative exists as a finite number on the boundary. **(c)** Near $(0,-3)$, $x+1 \neq 0$ and $q > 0$, so $f$ is smooth. Since $\nabla f(0,-3) = (-3,-1)$, $Df(0,-3)(h_1,h_2) = -3h_1 - h_2$. **(d)** The tangent plane is $z = 8 - 3x - (y+3) = 5 - 3x - y$. **(e)** The increment from $(0,-3)$ is $(-0.03,-0.01)$, so $f(-0.03,-3.01) \approx 8 - 3(-0.03) - (-0.01) = 8.10$.

12. Find the total differential of $f(x,y) = x/y$ wherever it exists, and justify that it is indeed the total differential.

**Solution.** The domain is $\{(x,y) : y \neq 0\}$. At $a = (a_1,a_2)$ with $a_2 \neq 0$, the candidate differential is the linear map
$$L(h) = \frac{h_1}{a_2} - \frac{a_1 h_2}{a_2^2}.$$
Direct subtraction gives the exact remainder
$$f(a+h) - f(a) - L(h) = \frac{h_2(a_1 h_2 - a_2 h_1)}{a_2^2(a_2 + h_2)}.$$
For $|h_2| < |a_2|/2$, its absolute value is at most $2(|a_1| + |a_2|)\,\|h\|^2 / |a_2|^3$. Dividing by $\|h\|$ gives a quantity tending to zero, proving $Df(a) = L$ at every point of the domain.

13. Determine whether the function

$$f(x,y) = \frac{x^3 y}{x^2+y^2}$$

can be extended so that it has a total differential at every point of $\mathbb{R}^2$. Find the total differential wherever it exists.

**Solution.** Yes: set $f(0,0) = 0$. For $r = \sqrt{x^2+y^2} > 0$,

$$|f(x,y)| = \frac{x^2|xy|}{x^2+y^2} \le \frac{x^2}{2} \le \frac{r^2}{2},$$

using $2|xy| \le x^2+y^2$. This proves continuity at the origin, and also $|f(h)-f(0)-0|/\|h\| \le \|h\|/2 \to 0$. Thus $Df(0,0) = 0$, and this is the only continuous extension. Away from the origin the denominator is nonzero, so the function is smooth. The quotient rule gives

$$Df(x,y)(h_1,h_2) = \frac{x^2y(x^2+3y^2)}{(x^2+y^2)^2}\,h_1 + \frac{x^3(x^2-y^2)}{(x^2+y^2)^2}\,h_2.$$

Hence the extension is differentiable at every point of $\mathbb{R}^2$.

14. Let $f(x,y) = -x^2-y^2+2x+4y-4$. Find a tangent plane $T$ to its graph that is perpendicular to the line $\{(t,t,t) \in \mathbb{R}^3 : t \in \mathbb{R}\}$. At which point does $T$ intersect the line $\{(0,0,t) \in \mathbb{R}^3 : t \in \mathbb{R}\}$?

**Solution.** At $(a,b,f(a,b))$, a normal vector to the tangent plane is

$$(-\partial_x f(a,b), -\partial_y f(a,b), 1) = (2a-2, 2b-4, 1).$$

The plane is perpendicular to the given line exactly when this normal is parallel to $(1,1,1)$. Its third coordinate forces the proportionality factor to be 1, so $a = 3/2$ and $b = 5/2$. At this point $f(a,b) = 1/2$ and $\nabla f(a,b) = (-1,-1)$. Therefore

$$T: \quad z - \frac{1}{2} = -\left(x-\frac{3}{2}\right) - \left(y-\frac{5}{2}\right), \qquad \text{or equivalently} \qquad x+y+z = \frac{9}{2}.$$

Setting $x = y = 0$ gives the intersection point $(0,0,9/2)$.